CSE 641 Computer Vision Project Report — Week 3

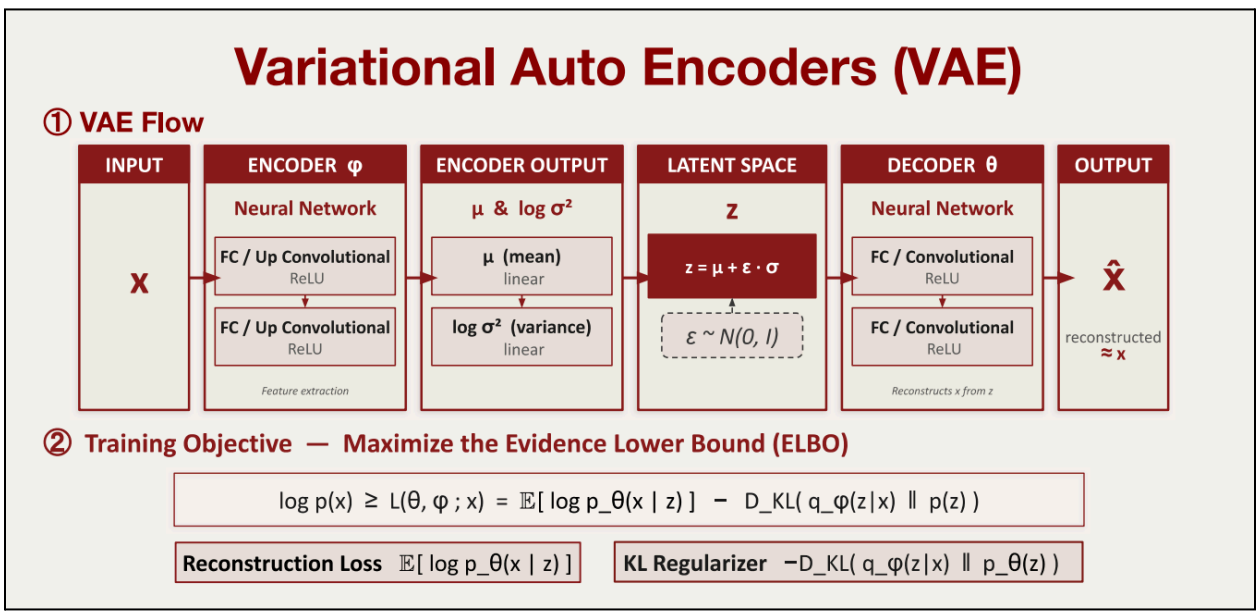
Project Title: Athletic Readiness: Data Generation

Group: Deep Vision

Introduction

In week 4, we focused on transitioning from data analysis to the modeling phase of the project. After completing preprocessing and EDA in week 3, the main objective of this week was to study and design generative models for synthetic athlete performance data generation. Our main focus was on understanding the architecture and working principles of **Variational Autoencoders (VAEs)** and **Generative Adversarial Networks (GANs)**, which will be used in the upcoming implementation phase.

Variational Auto Encoders (VAEs)



The VAE is a generative model which maps the input data x to a probability distribution in latent space rather than a fixed point that enables it to generate new realistic samples. The VAE has three components:

Part	Role
Encoder φ	Maps the input data x → (μ, log σ²); approximates posterior $q_\varphi(z\|x) \approx N(\mu, \sigma^2)$ with parameter φ.

Latent z	Sampled via $z = \mu + \varepsilon \cdot \sigma$, $\varepsilon \sim N(0,I)$ using the reparameterisation trick
Decoder θ	Reconstructs $\hat{x}$ from the z; learns generative model $p\theta(x\|z)$ with parameter θ.

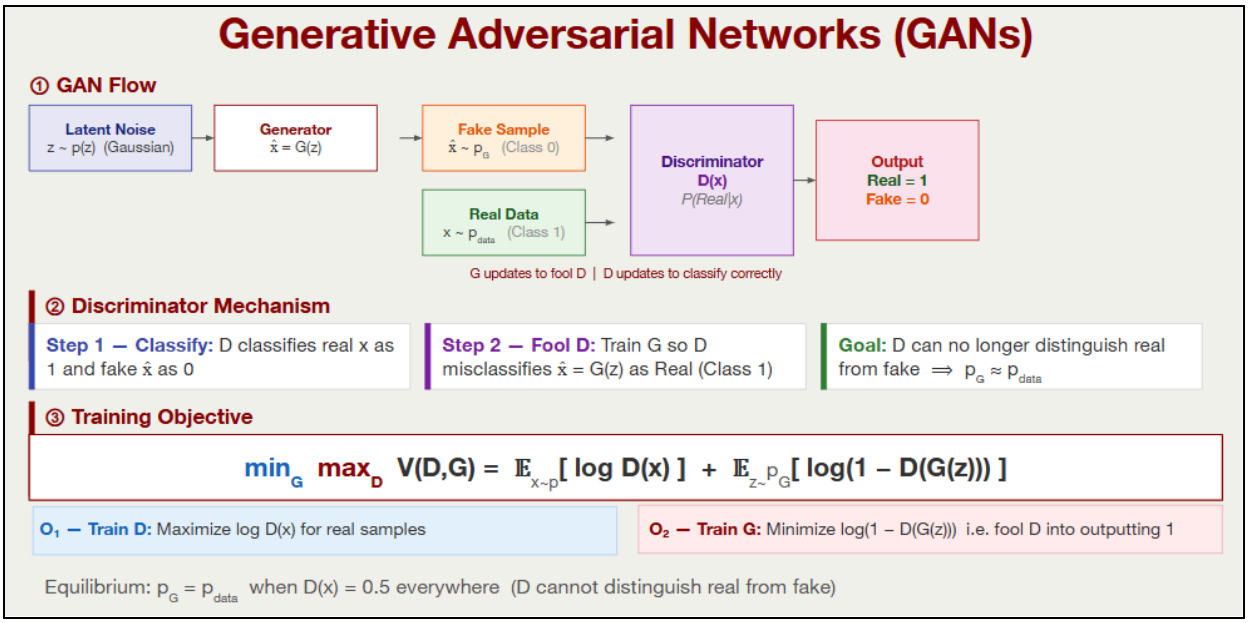
Training Objective (ELBO):
Here, the true marginal likelihood $p\theta(x) = \int p\theta(z)\, p\theta(x|z)\, dz$ is intractable, so the VAE maximises its variational lower bound (Eq. 3, Kingma & Welling):

$$L(\theta,\ \varphi;\ x) \ = \ -\ DKL(\ q\varphi(z|x)\ \|\ p\theta(z)\ ) \ + \ Eq\varphi(z|x)[\ log\ p\theta(x|z)\ ]$$

- The KL divergence term regularises the latent space, preventing the encoder from memorising inputs by keeping $q\varphi(z|x)$ close to the latent space with N(0,1).
- The reconstruction term ensures that the decoder faithfully reproduces all input features from sampled z which is as close as possible to the original input data.
- The reparameterisation trick ($z = \mu + \sigma \odot \varepsilon$, $\varepsilon \sim N(0,1)$) separates the randomness from learnable parameters which makes the entire pipeline differentiable and trainable end-to-end using backpropagation.

Why use VAE for Athletic Readiness Data Generation?
The sports analytics data is highly sensitive. The VAE encodes the input data into a probabilistic distribution which helps us preserve the privacy of the data. Also, the VAE learns a continuous latent distribution rather than a single point which maintains the original statistical integrity and correlation of the input data. VAEs have well-defined, stable training objectives that work directly on continuous tabular features. The KL Divergence term prevents the latent space from collapsing under the near deterministic feature relationships. It forces the encoder to learn a compressed, disentangled representation rather than a trivial identity mapping.

Generative Adversarial Network (GAN)



Key components studied:

- Generator: Produces fake samples from random noise.
- Discriminator: Tries to identify differences between the real and fake ones.
- Adversarial Training: Both models compete, improving data realism.

Training Objective (GAN):

Generative Adversarial Networks are trained using a min-max game between the generator and discriminator. The main training objective is to maximize the discriminator's loss and minimize the generator's loss. The discriminator tries to correctly classify real data and generated data while the generator tries to fool the discriminator by producing realistic synthetic samples.

At equilibrium, the generator learns the true data distribution and the discriminator becomes unable to distinguish real from fake samples (output ≈ 0.5).

Why use GAN for Athletic Readiness Data Generation?

GANs are powerful for generating highly realistic data distributions, making them suitable for replicating complex athlete performance patterns.

For this project:

- GANs can model sharp feature relationships, like the correlations between PTS, FGM, and MIN observed during EDA.
- GANs can generate high-quality samples, which makes synthetic data more similar to real athlete statistics.
- Since GANs do not reconstruct input data directly, they lower the risk of memorizing exact data and help protect privacy.

While model training we should make sure to check results as GAN can also face instability during training. Also, there can be risk of mode collapse and will require careful tuning.

Target for the next week

The following tasks are planned for the next phase:

VAE Implementation

- Build encoder and decoder using 3-layer MLP
- Implement reparameterization trick
- Train model on scaled dataset
- Generate synthetic samples

GAN Implementation

- Design generator and discriminator networks
- Train adversarial model
- Evaluate synthetic data quality

References:

Kingma, D. P., & Welling, M. (2022). *Auto-encoding variational Bayes* (arXiv:1312.6114). arXiv. https://arxiv.org/pdf/1312.6114

Goodfellow, I. J., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2014). Generative Adversarial Networks (arXiv:1406.2661). arXiv. https://arxiv.org/pdf/1406.2661